

Single Engine Aerodynamics

Critical Engine Definition: The engine whose failure would most adversely affect the performance or handling qualities of the aircraft. There are four factors that make the “Left engine” the critical in a conventional Multi Engine airplane.

Use the acronym PAST

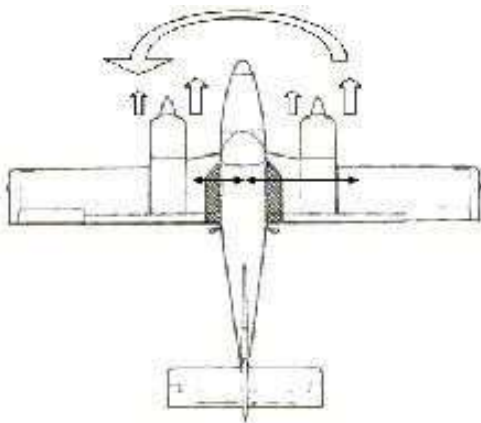
P- P Factor

A- Accelerated Slipstream

S- Spiraling Slipstream

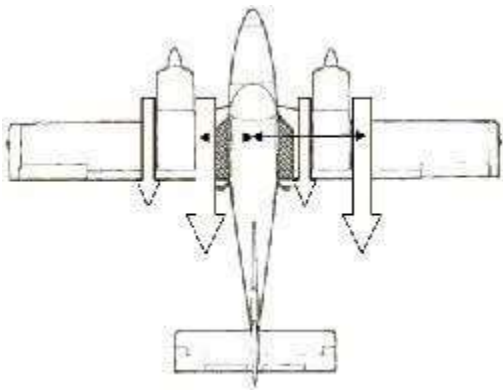
T- Torque Roll

- 1) **P FACTOR:** The descending blade produces more thrust than the ascending blade. If the left engine fails, the p-factor being produced from the right engine is farther from the longitudinal axis of the aircraft, creating more of a yawing moment



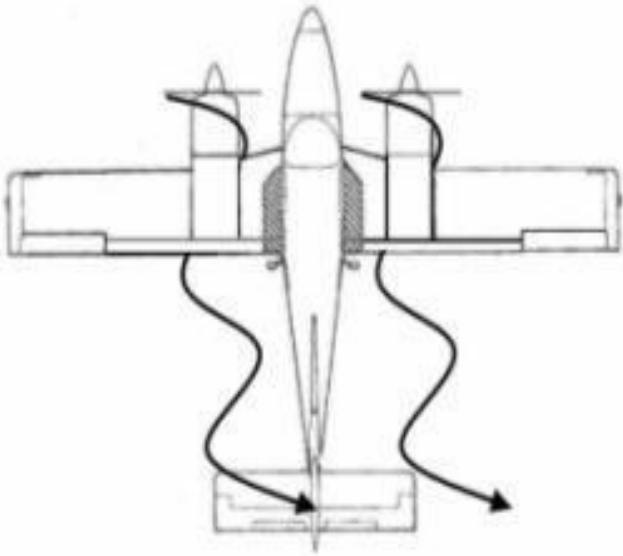
P-Factor

- 2) **ACCELERATED SLIPSTREAM**: The air being forced over the wing by the operating engine creates extra lift. The airflow from the Right Engine creates more induced airflow and creating a longer arm and rolling moment.



Accelerated Slipstream

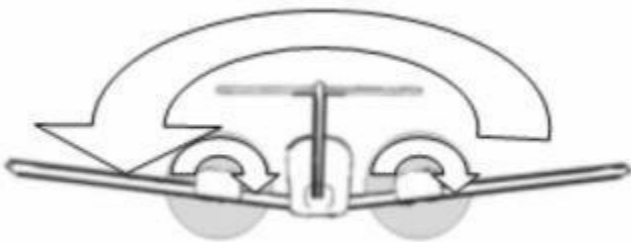
- 3) **SPIRALING SLIPSTREAM**: The effect of the propeller being pulled through the air while rotating creates a spiraling stream of air behind the propeller. This spiraling air from the left propeller, due to its direction of spiral, hits the rudder creating more airflow to make the rudder more effective. If the left engine fails, the spiraling slipstream from the right engine spirals away from the rudder. *The direction of spiral airflow is from High Pressure to Low Pressure, or Left to Right in the diagram above. The descending propeller blade is at a High angle of attack and produces Low Pressure air flow. Conversely the ascending propeller blade is at a Low angle of attack and produces High Pressure air flow. This results in the High Pressure airflow pushing the Low Pressure, spiraling slipstream, to the right.*



$H \rightarrow L$ $H \rightarrow H$

Spiraling Slipstream

- 4) **Torque**: The opposite reaction to the clockwise rotation of the propeller. Torque tries to roll the airplane opposite of the propeller's direction of rotation. If the left engine fails in a conventional Multi Engine airplane, torque tries to roll the aircraft to the left.



Torque Roll